

E-Commerce Sales Performance Analysis

Objective

The objective of this project is to analyze the Brazilian E-Commerce Public Dataset by Olist and identify patterns in sales performance, customer behavior, product trends, seller contribution, payment preferences, and geographic performance. The goal is to generate actionable business insights that support data-driven decision making.

Dataset Description

The dataset contains information related to:

- Customers
- Orders
- Order Items
- Products
- Product Categories
- Payments
- Reviews
- Sellers
- Geographic Information

Multiple datasets were merged using common keys such as `customer_id`, `order_id`, `product_id`, and `seller_id` to create a unified analytical dataset.

Data Cleaning Summary

The following data cleaning steps were performed:

1. Identified and handled missing values.
2. Removed duplicate records where applicable.
3. Corrected incorrect data types.
4. Converted date columns into datetime format.
5. Standardized column names and formats.
6. Verified data integrity after merging multiple datasets.
7. Documented unmatched records and merge validation checks.

Analysis Findings

Product Performance

- A small number of product categories generated the majority of revenue.
- Top-performing products contributed significantly to overall sales.

Geographic Performance

- Sales were concentrated in a limited number of cities and states.
- Certain regions consistently outperformed others in revenue generation.

Customer Behavior

- Repeat customers generated higher lifetime value.
- Purchasing activity showed clear seasonal patterns.

Payment Trends

- Credit card payments were the most commonly used payment method.
- Payment preferences varied across customer groups.

Seller Performance

- A small group of sellers contributed a large share of total revenue.
- High-performing sellers maintained stronger customer satisfaction scores.

Visualizations

The following visualizations were created:

1. Revenue by Product Category
2. Revenue by Region
3. Monthly Sales Trend
4. Payment Method Distribution
5. Review Score Distribution
6. Correlation Heatmap
7. Top Products by Revenue
8. Top Sellers Analysis

These visualizations helped identify trends, patterns, and business opportunities within the dataset.

Final Conclusion

The analysis revealed that revenue is concentrated among a limited number of product categories, regions, and sellers. Customer retention plays an important role in overall business value, while payment methods and review scores provide additional insight into customer behavior.

Recommendations

- Increase marketing investment in high-performing product categories.
- Expand operations in top-performing regions.
- Develop loyalty programs to increase repeat purchases.
- Support high-performing sellers through strategic partnerships.
- Monitor customer reviews to improve product quality and service experience.

By applying these recommendations, the company can improve revenue growth, customer satisfaction, and long-term business performance.